

Practical Astronomy Intensive

Four days for quiz-selected students who know the theory and have never used a telescope. Sky reading, telescope handling, safe solar observing and a first taste of astrophotography — book astronomy turned into finding things in a real sky.

9	52	19 h 30 min	11 h	Amateur Astronomy Club
SESSIONS	CONCEPTS TAUGHT	TOTAL	OUTDOORS	AUDIENCE

Moon: start when the moon is **0.5–2.5 days old** (🌑 new → 🌒 waxing crescent)

ASSUMED ALREADY KNOWN

The Solar System Family

The Sun is a Star

The Magnitude Scale

Constellations

Galaxies

Structure of the Milky Way

Nebulae & Star Clusters

Messier & NGC Catalogues

Lives of Stars

Lenses, Mirrors & Focal Length

Moon Orbits Earth

Tidal Locking

Solar Eclipses

Lunar Eclipses

Ecliptic & Zodiac

The Heliocentric Model

Planets: the Wanderers

Spectrum & Colour

Scale of the Solar System

Latitude & Longitude

The Ocean of Air

How We See

Hot & Cold: Temperature

Big Numbers & Powers of Ten

Maps & Scale Models

Models in Science

Cardinal Directions

Closed under prerequisites: 61 concepts in total are taken as prior knowledge.

1 Day 1 — Where things are: the celestial sphere

CLASSROOM · 2 H · DAY 1

KIT Projector · Star chart / planisphere · Paper, pencils, protractor

L2 Angles on the Sky

Sky positions and sizes are measured in angles: your fist at arm's length spans about 10° , the Moon half a degree. Degrees, arcminutes and arcseconds are the astronomer's ruler.

L2 Diurnal Motion

The whole starry sky wheels westward through the night, circling a fixed point — the celestial pole. This is Earth's rotation seen from the inside.

L2 The Pole Star

Polaris (Dhruva) sits nearly at the north celestial pole, so it barely moves and always marks north. Its altitude above the horizon equals your latitude.

L3 Celestial Sphere

A model sphere of infinite radius carrying the stars, with celestial poles and a celestial equator projected from Earth's own. It turns star positions into geometry we can compute.

L3 Altitude–Azimuth Coordinates

The local way to point: azimuth around the horizon from north, altitude up from the horizon. Simple and intuitive — but every star's alt-az changes by the minute.

L3 Circumpolar Stars

Stars close enough to the celestial pole never set for your latitude — they circle it all night, all year. Which stars are circumpolar depends only on where you stand.

L3 Sky from Different Latitudes

At the equator every star rises and sets vertically; at the poles nothing rises or sets at all. Your latitude completely decides which sky you own — Canopus for Chennai, none for London.

L3 Meridian & Transit

The meridian is the north–south line through your zenith; objects are highest — and best observed — when they cross (transit) it. Transit timing is the backbone of classical positional astronomy.

TRAINER NOTE They can define the celestial sphere and cannot point at the meridian. Do this indoors first, then make every claim checkable outside tonight. Angular measure comes first because every other session leans on it and nobody arrives with it — the quiz never asked them to measure anything.

SKY needs **Dark sky (no moon)** · Constellations, Bright stars, Milky Way, Horizon / cardinal points

KIT Star chart / planisphere · Red torch

L1 Stars Differ

Stars are not identical dots: they differ in brightness and in colour — steely blue Vega, golden Arcturus, red Betelgeuse. These differences are real physics, visible to the naked eye.

L2 Light Pollution

Artificial skyglow erases faint stars; the Bortle scale grades sky darkness from 1 (pristine) to 9 (inner city). Shielded, warm, dimmed lighting brings stars back.

L2 The Milky Way Band

From a dark site a faint luminous band arches across the sky — the combined light of billions of stars in our galaxy's disc, seen from inside it.

L2 The Eye as a Detector

Cone cells see colour but need light; rod cells see faint things only in grey. That's why nebulae look ghostly-grey in the eyepiece yet glow red and blue in photographs.

L1 Asterisms

Popular star patterns that aren't official constellations — the Big Dipper is part of Ursa Major, the Summer Triangle spans three constellations. The sky's informal landmarks.

L2 Star Names & Bayer Letters

Bright stars carry ancient proper names (many Arabic: Aldebaran, Deneb); astronomers label stars α , β , γ by brightness within each constellation. Two naming systems, one sky.

L2 Dark Adaptation & Averted Vision

Eyes take 20–30 minutes to reach full night sensitivity, and one white-light glance resets them. Looking slightly beside a faint object uses your more sensitive off-centre retina.

L2 Star Charts & Planetarium Apps

Reading a sky map: orientation, scale, magnitudes, and matching chart to sky. A planisphere or app turns the catalogue sky into 'what's above me right now'.

TRAINER NOTE No telescopes tonight, on purpose. Twenty minutes of dark adaptation, ruined by one white phone screen — let that happen once and the red-torch rule needs no further explanation. A moonless night is not a compromise here, it is the point: they have read about the Milky Way and most have never seen it.

3 Day 2 — How a telescope actually works

CLASSROOM · 2 H 30 MIN · DAY 2

KIT Dobsonian reflector · Refractor · Eyepiece set · Projector

L3 Seeing & Transparency

Seeing is atmospheric steadiness (how much stars boil); transparency is clarity (how faint you can go). A hazy still night can beat a crisp windy one for planets.

L3 Refractors & Reflectors

Refractors use a lens, reflectors a mirror, catadioptrics both. Each design trades cost, size, and image quality differently — there is no single 'best telescope'.

L3 Aperture & Light Grasp

Aperture is the diameter of the main lens or mirror — it decides how faint you can see and how fine you can resolve. Aperture, not magnification, is a telescope's true power.

L3 Magnification & Eyepieces

Magnification = telescope focal length ÷ eyepiece focal length; swap eyepieces to change power. More is rarely better — the atmosphere and aperture set a useful ceiling.

L3 Focal Ratio & Field of View

f-ratio (focal length ÷ aperture) sets how 'fast' the optics are and how wide a field you get. Slow scopes suit planets; fast ones frame star fields and nebulae.

L4 Exit Pupil & Eye Relief

The beam leaving the eyepiece should match your night-opened pupil (~7 mm young, ~5 mm older) — the physics behind why 7×50s suit youngsters and 8×32s suit elders.

L4 Resolution & the Airy Disc

Diffraction blurs every star into a tiny Airy disc; aperture sets the smallest detail resolvable (Dawes' limit). Why splitting close double stars needs inches, not power.

TRAINER NOTE Hands on the actual instruments while explaining. The '600x!' box-claim misconception dies fastest when they compute exit pupil for themselves and then look through it.

SKY needs **Dark sky (no moon)** · Double stars, Jupiter, Saturn, Open clusters, Globular clusters, Nebulae, Galaxies

KIT Dobsonian reflector · Binoculars · Eyepiece set · Star chart / planisphere · Red torch · Paper, pencils, protractor

L2 Binocular Astronomy

A 7×50 binocular — 7× magnification, 50 mm objectives — shows Moon craters, Jupiter's moons and hundreds of clusters. The best first instrument, and many observers' favourite forever.

L2 First Light: Using a Telescope

Align the finder in daylight, start with the lowest power, centre on something bright, focus slowly. Ten minutes of drill separates a magical first night from a frustrating one.

L2 Observing Logs & Sketching

Date, time, sky conditions, instrument, and what you actually saw — a log turns stray sessions into a record of growth, and sketching trains the eye to see more.

L3 Star Hopping

Navigating from a bright star to a faint target through chart-matched patterns, hop by hop. The craft skill that turns a chart reader into an observer.

L3 Deep-Sky Observing

Hunting clusters, nebulae and galaxies — faint, extended targets that want aperture, dark skies, low power and dark-adapted eyes. The 'faint fuzzies' that hooked every amateur.

L3 Planetary Observing

Jupiter's belts, Saturn's rings, Mars at opposition: small, bright targets that reward high power, steady seeing and a patient eye at the eyepiece.

L4 Double & Variable Stars

Coloured double stars test optics and delight the eye; variable stars change brightness on the clock. Both offer structured observing programmes with real scientific value.

L4 Visual Nebula Filters

UHC and OIII filters pass only the wavelengths nebulae emit, dumping skyglow — the Veil Nebula appears from nothing. They help emission nebulae only; galaxies get no rescue.

TRAINER NOTE First light on doubles and planets — bright, easy, forgiving, and the fastest route to a beginner actually finding something. Then spend the dark half on deep sky while the sky is at its blackest. Manage the disappointment honestly and early: M31 is a grey smudge, not the photograph. Averted vision is the skill of the night. They have read the Messier table — tonight they find three of them by star-hopping, with no GoTo.

5 Day 3 — Distances, and how anyone knows

CLASSROOM · 2 H · DAY 3

KIT Projector · Paper, pencils, protractor

L3 The Inverse-Square Law

Double the distance, quarter the light — brightness falls as $1/d^2$. This single rule connects a star's true power, its distance, and how bright it looks to us.

L3 Star Colours & Temperature

A star's colour is a thermometer reading: red ~3,000 K, yellow ~6,000 K, blue-white 10,000 K+. Hotter objects glow bluer — physics you can check with your own eyes.

L4 Airmass & Extinction

Light through more atmosphere gets dimmed and reddened — why the horizon Sun is orange and why photometry corrects for 'airmass'. Observe high, image high.

L3 Light Years & Cosmic Scale

The nearest star is 4.2 light years away — 7,000× farther than Neptune. Grasping the jump from solar-system scale to interstellar scale is a rite of passage.

L4 Stellar Parallax

Nearby stars shift minutely against the background as Earth orbits — the parallax that defines the parsec and anchors every distance in astronomy.

L4 Luminosity & Absolute Magnitude

Apparent brightness mixes true output with distance; absolute magnitude removes distance (all stars imagined at 32.6 ly). Only then can stars be honestly compared.

TRAINER NOTE They memorised that Sirius is 8.6 light-years away. Ask how that number was obtained. Parallax with a thumb at arm's length, then the arithmetic — this is the session that turns a fact back into a measurement.

6 Day 3 — The Moon arrives

OBSERVING NIGHT · 2 H 30 MIN · DAY 3

SKY needs **Thin crescent** · The Moon, Double stars

KIT Dobsonian reflector · Binoculars · Eyepiece set · Smartphone · Phone-to-eyepiece adapter · Red torch · Paper, pencils, protractor

L2 Craters & Maria

The Moon's dark 'seas' are ancient lava plains; its craters are asteroid scars, preserved for billions of years with no air or water to erase them. The face of the Moon is a history book.

L2 Lunar Observing

Craters, maria, mountain shadows — the Moon is best along the terminator, where low sunlight throws relief into drama. Full moon is actually the flattest, worst time.

L2 Smartphone Astrophotography

A phone held (or bracketed) at the eyepiece captures the Moon and planets; night modes catch constellations and even the Milky Way. The zero-cost gateway to imaging.

L4 Lunar Libration

The Moon rocks gently side-to-side and up-and-down as it orbits, letting us peek around its edges — 59% of the surface is visible over time, not just 50%.

TRAINER NOTE The moon is deliberately last. The camp starts at new moon so the two dark nights come first, and by tonight a crescent has climbed into the evening sky — the sky supplies the lesson on schedule. Work the terminator: craters are invisible at full moon and unmissable along the day–night line. Sketch first — a pencil forces them to actually look — THEN bring out the phone adapter: the moon is the one target where a phone at the eyepiece gives a genuinely good photograph on the first try, and everyone goes home with one.

7 Day 4 — The Sun, safely

SOLAR SESSION · 1 H 30 MIN · DAY 4

SKY needs **Any moon** · The Sun

KIT Refractor · Solar filter (front aperture) · Paper, pencils, protractor

L3 Anatomy of the Sun

Outwards from a 15-million-kelvin fusion core: a radiation zone, a boiling convection zone, then the photosphere we call the surface — mottled like a pot of cooking oatmeal — the thin chromosphere, and the corona. The corona is the puzzle: at millions of degrees it is hundreds of times hotter than the surface beneath it, and nobody has fully explained why.

L3 Sunspots & Solar Activity

The Sun's magnetic weather: spots, flares and mass ejections rising and falling on an 11-year cycle. It drives aurorae and can rattle satellites and power grids.

L3 Solar Observing (Safely)

NEVER look at the Sun without certified filters — projection and proper solar film show sunspots safely; hydrogen-alpha scopes reveal prominences. Safety protocol is the first lesson.

TRAINER NOTE Front-aperture filter fitted and checked by the instructor personally, every time, before anyone's eye goes near it. Never an eyepiece filter. State the rule, then explain what a Herschel wedge does to an unfiltered finder — the fear is the lesson.

8 Day 4 — Indian sky, and what to do next

CLASSROOM · 2 H · DAY 4

KIT Projector · Star chart / planisphere

L3 Tithi: the Lunar Day

A tithi is the time the Moon gains 12° on the Sun — 30 tithis from new moon to new moon. Being angle-based, tithis stretch and shrink; a date can even skip one.

L2 Indian Star & Constellation Names

Dhruva (Polaris), Saptarshi (the Big Dipper), Arundhati–Vasishtha (the Mizar–Alcor pair), Rohini (Aldebaran), Chitra (Spica): the Indian sky map atop the same stars.

L3 Citizen Science

Variable-star estimates (AAVSO), occultation timings, meteor counts, comet hunting: organised programmes where amateur measurements feed real research.

L3 The 27 Nakshatras

The ecliptic divided into 27 lunar mansions of $13^\circ 20'$ each — one for each day of the Moon's sidereal circuit, each anchored to a marker star (yogatara).

L3 Panchanga: Five Limbs

The traditional almanac's five daily elements — vara (weekday), tithi, nakshatra, yoga, karana — each a different function of Sun and Moon longitudes. An ephemeris in cultural form.

TRAINER NOTE The quiz is named for Aryabhata and its study material never mentions the nakshatras — close that gap here. Saptarishi is the Big Dipper they found on night one; Arundhati–Vasishtha is a naked-eye double they can still check. Finish with citizen science so the camp has an afterwards.

SKY needs **Thin crescent** · Milky Way, Star trails, Constellations

KIT Smartphone · DSLR / mirrorless · Tripod · Red torch · Power bank / batteries

L3 Exposure: Aperture, Shutter, ISO

Three controls set how much light a camera records and how noisy it looks. At night you run all three near their limits — astrophotography is exposure at the extreme.

L3 Nightscape Photography

Wide-angle sky-over-landscape shots from a fixed tripod. The NPF/500 rule caps exposure before stars trail — beyond it, you need tracking.

L3 Star Trail Photography

Stack hundreds of fixed-tripod frames and stars draw arcs around the celestial pole — diurnal motion made visible and beautiful in one image.

TRAINER NOTE The taste of astrophotography, not the meal. The day-4 crescent sets in the first hour — watch it go down, then the sky is dark and every phone and camera comes out on a tripod. One wide nightscape and one short star-trail stack each; the trail arcs centre on the pole star they found on night one, which closes the camp's circle. Anyone hungry for more gets pointed at the full astrophotography workshop.

FULL KIT LIST

Binoculars	sessions 4, 6
Dobsonian reflector	sessions 3, 4, 6
DSLR / mirrorless	sessions 9
Eyepiece set	sessions 3, 4, 6
Smartphone	sessions 6, 9
Phone-to-eyepiece adapter	sessions 6
Power bank / batteries	sessions 9
Projector	sessions 1, 3, 5, 8
Red torch	sessions 2, 4, 6, 9
Refractor	sessions 3, 7
Solar filter (front aperture)	sessions 7
Star chart / planisphere	sessions 1, 2, 4, 8
Paper, pencils, protractor (<i>consumable</i>)	sessions 1, 4, 5, 6, 7
Tripod	sessions 9